

JOINT SESSION · WEEK 1

Kick-off

Comparing Europe with Data. Meet the people you will work with, understand how the COIL works and hold your first team meeting.

Week 1 · 12 Oct – 18 Oct 2026 · 45 minutes · Microsoft Teams



What we will do today



5 min Welcome from the teachers of the three universities. What is a COIL and what is Ulysseus.

10 min How the COIL works: this website, the schedule, three deliverables, the dataset, extra points and certificate.

25 min First team meeting in break-out rooms (teams announced by the teachers, with students from the three universities): introduce yourself with your fact about your country, choose a communication channel and a weekly meeting time.

5 min Questions and closing.

WELCOME

Your teaching team



Daniel Ruiz Romera

University of Seville

druiz8@us.es



Alena Mojsejová

Technical University of Košice

alena.mojsejova@tuke.sk



Bojan Pejović

University of Montenegro

bojan.p@ucg.ac.me



Dana Paľová

Technical University of Košice

dana.palova@tuke.sk



**Víctor Ernesto Pérez
León**

University of Seville

vpleon@us.es

The COIL is designed and run jointly by teachers from the three universities. We share the schedule, the dataset and the final presentation, and each of us guides and grades the students of our own university.

WHAT IS THIS?

A COIL, inside the Ulysseus alliance



COIL

Collaborative Online International Learning: your course is connected with courses at partner universities, and you work online in international teams.



Ulysseus

A European University alliance. The University of Seville, the Technical University of Košice and the University of Montenegro are members.



This COIL

Nine weeks, three countries, one data project: compare Spain, Slovakia and Montenegro with the rest of Europe using real data.

One question, real data, three countries

1

Question

A socio-economic or policy question

2

Data

COIL dataset: 32 European countries

3

Analysis

Regression, classification, PCA, clustering

4

Dashboard

For a non-specialist audience

5

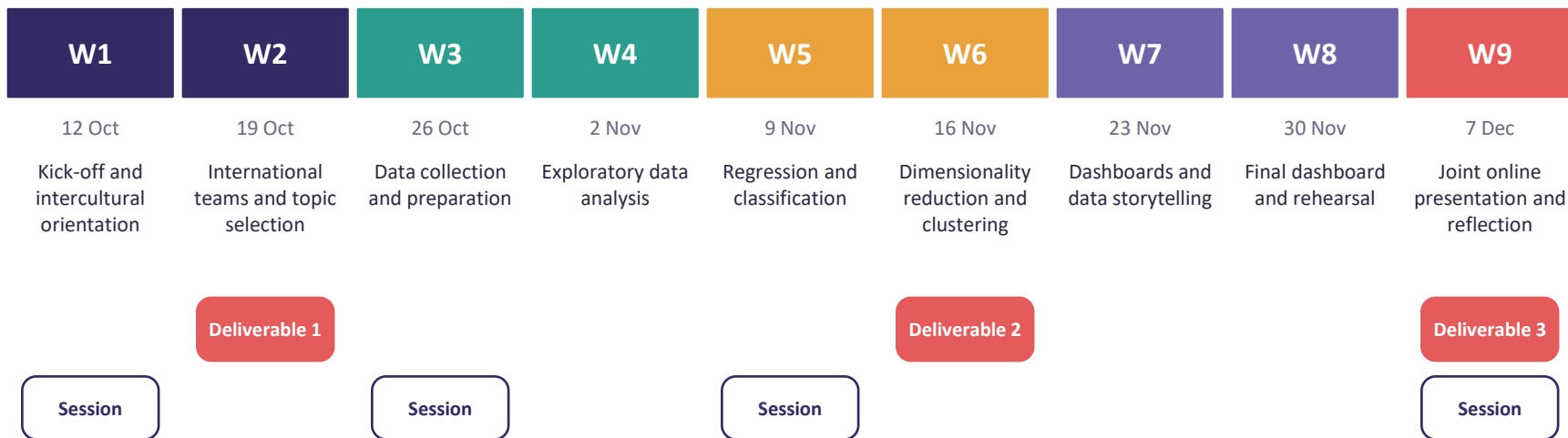
Presentation

Week 9, to all three universities

Example projects are on the website (Projects): education and youth unemployment, the digital divide, ageing and public spending, women in the labour market, the green transition, inequality. You can also propose your own question.

SCHEDULE

Nine weeks · 12 October – 11 December 2026



Phases: orientation · data · methods · communication · presentation. Dates are provisional until confirmed by the three universities; the website always shows the current version.

Three deliverables, always Sunday, 23:59 CET

1

Week 2
Sunday 25 October

Team charter and topic proposal

Who: Team

Format: One PDF, maximum 2 pages (use the templates in the guide)

2

Week 6
Sunday 22 November

Analysis notebook

Who: Team

Format: Python or R notebook (.ipynb, .Rmd or Quarto), or Gretl script (.inp) with its output, + exported HTML or PDF + your dataset (CSV or Excel)

3

Week 9
Sunday 13 December

Final dashboard, presentation and final reflection

Who: Team (dashboard, slides and presentation) + individual (reflection)

Format: Dashboard link or PDF + slides (maximum 6 slides, 5-minute presentation) + one-page PDF reflection

Submit with the Submit button of each deliverable on the website. The full list of what each deliverable must include is in the schedule and the student guide.

Deliverable 1 • Team charter and topic proposal

It must include

- Team charter: team name, members and universities, roles, communication channel, weekly meeting time, how you take decisions, what you do if someone does not contribute.
- Topic proposal: research question, why it matters for Spain, Slovakia and Montenegro, 3–6 indicators from the COIL dataset (with their codes), countries and years, and the methods you expect to use.
- A short note on your use of AI tools (which tool, for what, and how you checked the result), or a sentence saying you did not use any.

Format

One PDF, maximum 2 pages (use the templates in the guide)

Templates

Topic proposal and team charter templates are in the Guide section of the website.

The COIL dataset, ready to use

32

European countries, plus the EU aggregate

21

indicators: labour, economy, public finance, population, education, digital, environment

16

years (2010–2025), one row per country and year

3

file formats: long, wide (Excel, Power BI) and Gretl

Sources and gaps

- Sources: World Bank (World Development Indicators) and IMF (World Economic Outlook). Each indicator keeps its code and unit.
- Some series have gaps, especially for Montenegro. The data workshop (week 3) shows how to deal with them.
- Download everything from the Data section of the website.

TOOLS

Use the tool of your course

Python

pandas, statsmodels, scikit-learn;
Jupyter notebook

R

lm, glm, prcomp, kmeans, fixest,
ggplot2

Gretl

Free econometrics package:
regression, logit, panel, PCA

Power BI

Dashboards; clustering in scatter
charts; Python or R visuals



AI tools are allowed

On two conditions: you declare in every deliverable which tool you used and for what, and you check everything it produces. All figures must come from the data, never from the AI.

The methods module shows every method step by step in the four tools.

ONE PLACE FOR EVERYTHING

The COIL website is the reference



Session links

and the vote on dates and times



Submissions

Submit button for each deliverable



Data

CSV files, indicators and coverage



Materials

Slides, methods module and code



Guide

Rules, templates and contacts

If something in a slide or a message differs from the website, the website is right.

WHY IT IS WORTH IT

Extra points and a certificate

+1

/ 10

University of Seville

+10

/ 100

Technical University of Košice ·
University of Montenegro

Extra points on your course grade for students who complete the final presentation.



Certificate

Certificate of participation in a Ulysseus European University COIL, signed by three European universities: Seville, Košice and Montenegro. Add it to your CV and LinkedIn.

Your first team meeting

1

Introduce yourself: name, university, what you study, and your surprising fact about your country.

2

Choose one communication channel for the team (and agree to answer within 48 hours on working days).

3

Fix a weekly meeting time. All three countries are in the same time zone (CET).

4

Exchange first ideas for your question: look at the example projects on the website.



25

minutes, then back to the main
room

BEFORE NEXT WEEK

Your checklist for week 2

- ☐ Read the student guide and the schedule on the website.
- ☐ Hold your first full team meeting and write your team charter.
- ☐ Choose your question and 3–6 indicators from the COIL dataset.
- ☐ Submit deliverable 1 on the website by Sunday 25 October, 23:59 CET.
- ☐ Next joint session: Data workshop, week 3. Vote for the date and time on the website.

Questions?